# Practice workbook · Name thorax and airway structures and the route of air

Trachea, bronchi, bronchioles, alveoli, ribs, intercostal muscles, sternum, diaphragm on diagrams and models.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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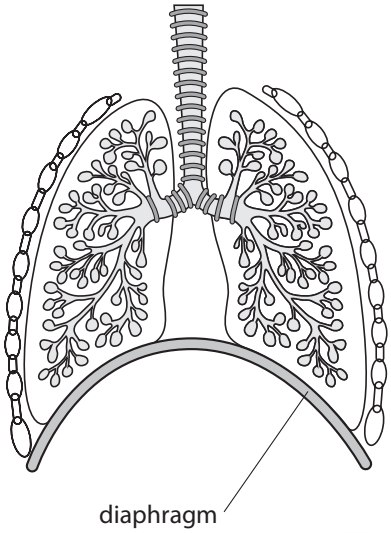
Type ID: HBI-THORAX-STRUCTURE-NAME

# 1. Question 4 · part a.i · 3 marks

June 2024 Paper 2

Worked-example retry · Attempt part a.i only.

**4** (a) The diagram shows the human lungs.




(i) Add the following labels on the diagram

(3)

- trachea
- bronchus
- alveoli

(ii) Give four features of the lungs that adapt them for gas exchange.

(4)

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## 1. Question 4 · part a.i · 3 marks · Working and answer

Working and answer

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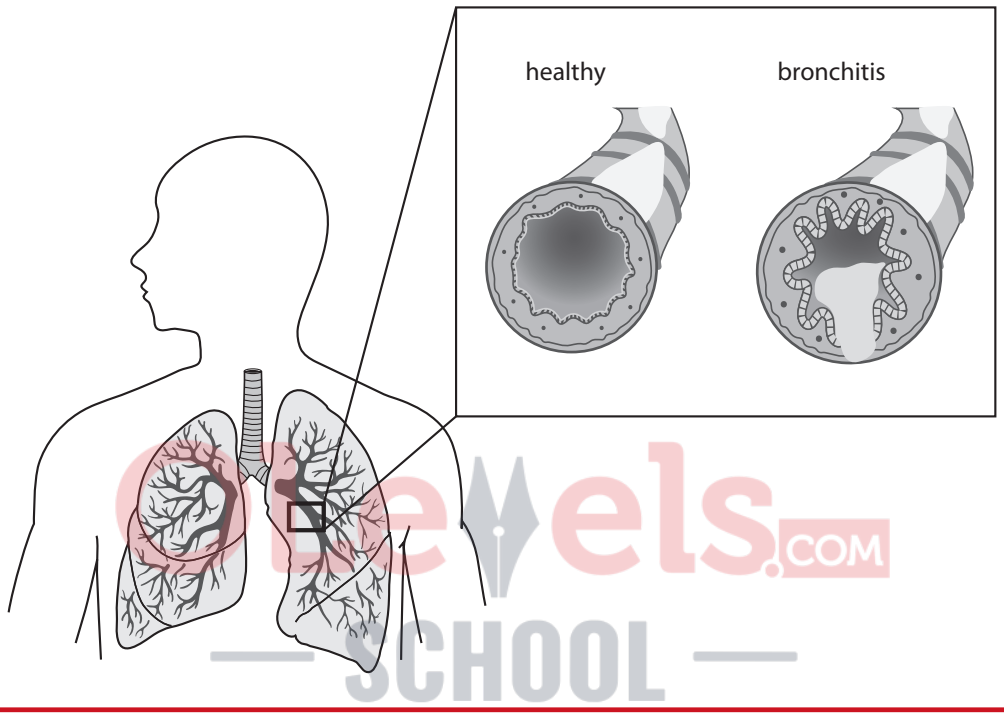
## 2. Question 6 · part a · 1 mark

January 2020 Paper 1

Further practice · Attempt part a only.

**6** Chronic obstructive pulmonary disease (COPD) is a group of lung diseases that includes bronchitis and emphysema.

The diagram shows how bronchitis affects the lungs.




(a) Name the part of the lungs affected by bronchitis.

**(1)**

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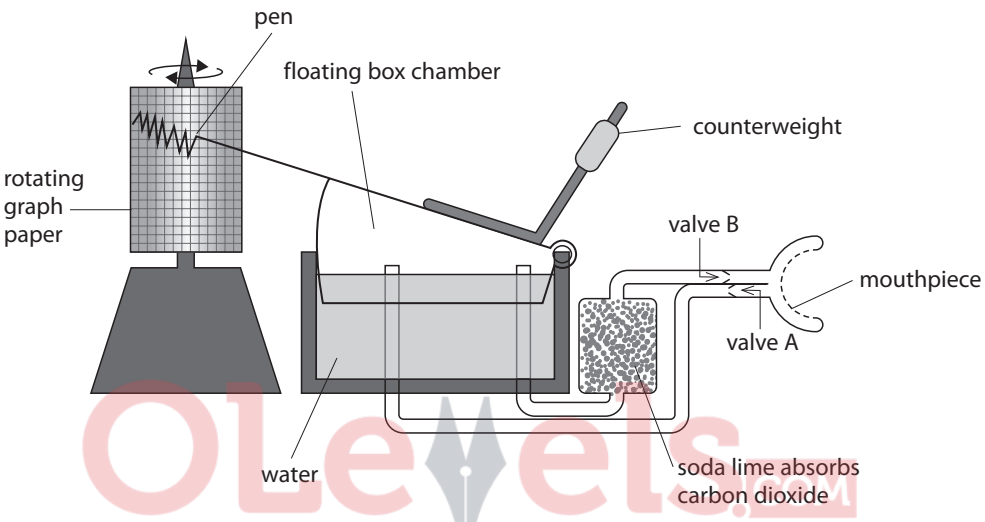
## 2. Question 6 · part a · 1 mark

January 2020 Paper 1

Further practice · Attempt part a only.

(b) The diagram shows a spirometer.

A spirometer measures how quickly a person breathes in and out.




(i) Describe how a spirometer is used to obtain a trace of a person's breathing pattern.

**(4)**

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## 2. Question 6 · part a · 1 mark · Working and answer

Working and answer

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# 3. Question 2 · part a.iii · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iii only.

**2** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The breathing system consists of airways that allow the movement of air into the lungs. Air can be breathed in through the nose or through the mouth. The lining of the nose has a good supply of blood capillaries and a layer of sticky mucus.

The whole of the trachea and the larger air passages have C-shaped rings of cartilage within their walls. These rings help the air to flow freely into the lungs. The trachea divides into two tubes which then divide further into finer branches. These finer branches eventually end in air sacs. It is in the air sacs that gaseous exchange occurs.

The lungs are located in the thorax which is surrounded by the ribs. The thorax is separated from the abdomen by the diaphragm. Together, the ribs and the diaphragm can change the volume of the thorax and therefore the pressure in the thorax.

During breathing, at rest, an average of 0.50 $dm^3$ of air is breathed in and out. The air breathed in contains 20% oxygen and the air breathed out contains 16% oxygen. At rest, an adult will breathe 17 times a minute on average. The breathing rate is controlled by a part of the brain which receives information about carbon dioxide levels in the blood.

Carbon monoxide may be present in the air that is breathed in. The carbon monoxide can enter the blood and combine with haemoglobin in the red blood cells. Carbon monoxide is found in cigarette smoke and car exhaust fumes.

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(a) (i) Discuss whether it is better to breathe in through the nose or the mouth. (lines 2 to 3)

**(3)**

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# 3. Question 2 · part a.iii · 1 mark

January 2022 Paper 2

Further practice · Attempt part a.iii only.

(ii) Explain why the larger air passages have C-shaped rings of cartilage within their walls. (lines 4 to 5)

**(2)**

(iii) The trachea divides into two tubes.

What is the name of these tubes? (line 6)

**(1)**

- ☒ **A** alveoli
- ☒ **B** bronchi
- ☒ **C** bronchioles
- ☒ **D** oesophagus

(b) Which of these is the name of the part of the brain that controls breathing rate? (line 16)

**(1)**

- ☒ **A** cerebrum
- ☒ **B** cerebellum
- ☒ **C** hypothalamus
- ☒ **D** medulla oblongata

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## 3. Question 2 · part a.iii · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part a.i · 2 marks

June 2023 Paper 1

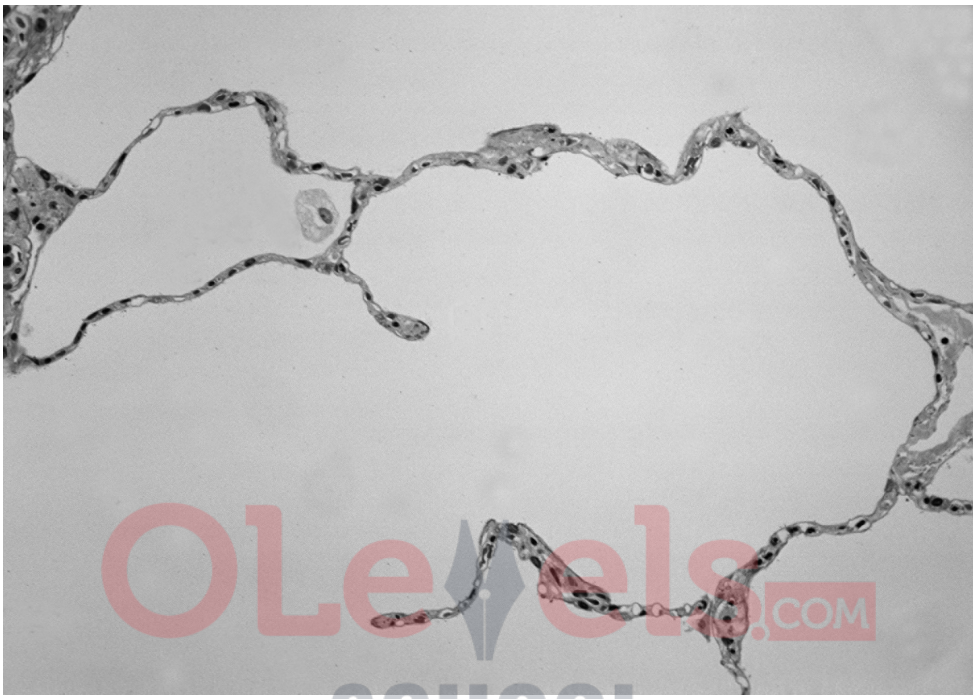
Further practice · Attempt part a.i only.

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**2** The photograph shows alveoli in the lungs. Alveoli are the site of exchange of gasses between the air and the blood.



(Source: © Ed Reschke/Getty Images)

(a) (i) State, in the correct order, the structures that air must pass through from the atmosphere to the alveoli.

**(2)**

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.....

(ii) State two features, shown in the diagram, that allow the efficient exchange of gases.

**(2)**

1 .....

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2 .....

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## 4. Question 2 · part a.i · 2 marks · Working and answer

Working and answer

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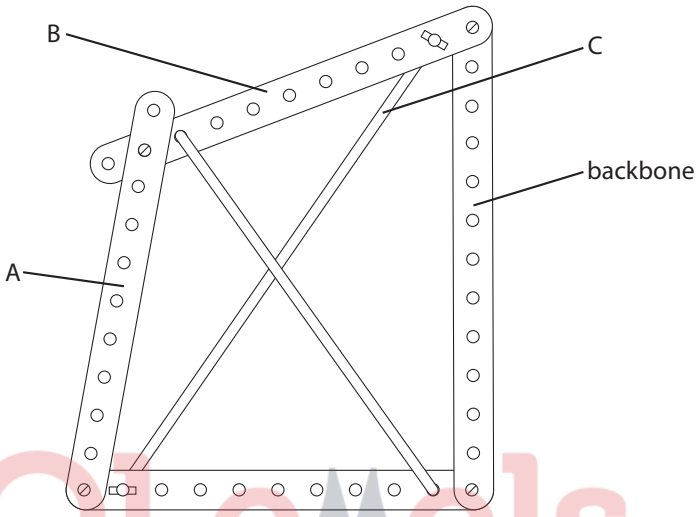
# 5. Question 5 · part a.i · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.i only.

**5** (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.




(i) State which parts of the breathing system are represented by A, B and C.

**(3)**

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

**(3)**

1 .....

2 .....

3 .....

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## 5. Question 5 · part a.i · 3 marks · Working and answer

Working and answer

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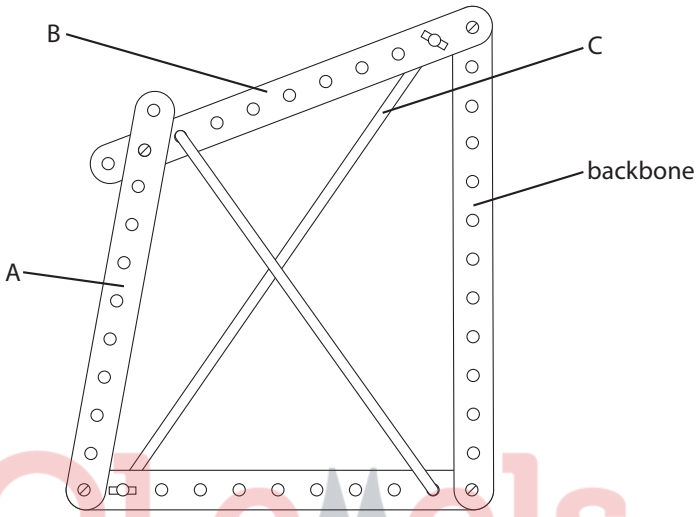
# 6. Question 5 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

**5** (a) The diagram shows a model that can be used to demonstrate the process of breathing.

The part that represents the backbone has been labelled.




(i) State which parts of the breathing system are represented by A, B and C.

**(3)**

A .....

B .....

C .....

(ii) State three other structures, not shown in the diagram, that are needed for breathing in the body.

**(3)**

1 .....

2 .....

3 .....

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## 6. Question 5 · part a.ii · 3 marks

June 2025 Paper 1

Further practice · Attempt part a.ii only.

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(iii) Describe how the structures present in the thorax are used to allow a person to inhale.

**(6)**

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(b) Give the balanced chemical symbol equation for the process of aerobic respiration.

**(3)**

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**(Total for Question 5 = 15 marks)**

## 6. Question 5 · part a.ii · 3 marks · Working and answer

Working and answer

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# 7. Question 4 · part b.i · 1 mark

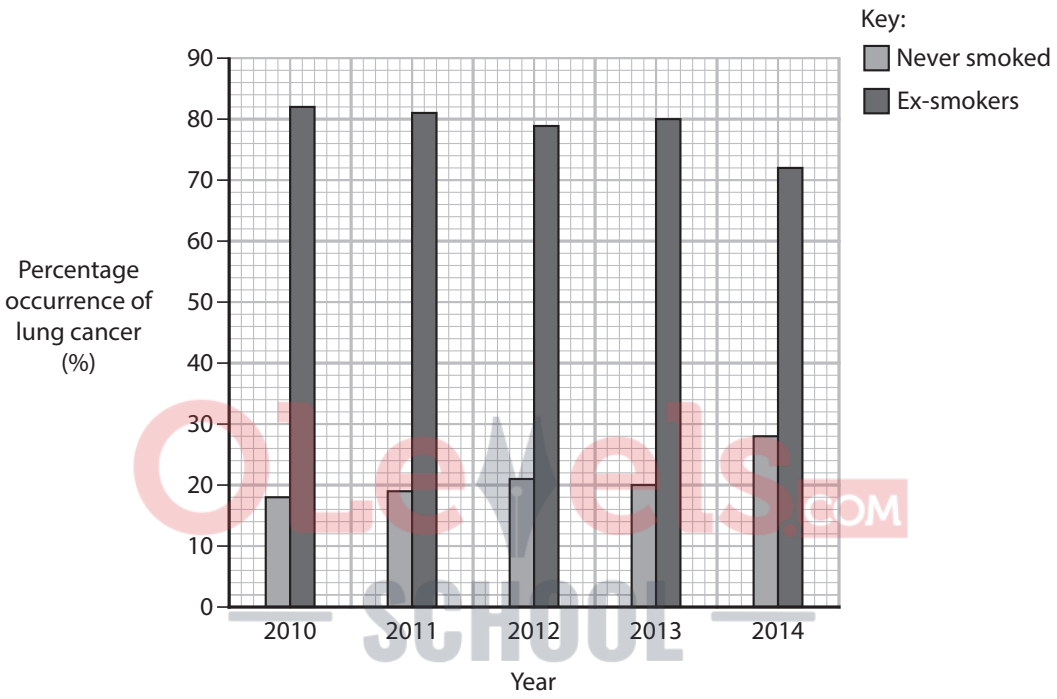
June 2025 Paper 2R

Further practice · Attempt part b.i only.

**4** (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.




(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

**(2)**

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## 7. Question 4 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

**(2)**

ratio = ..... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

**(2)**

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(iv) Name the substance in cigarette smoke that causes lung cancer.

**(1)**

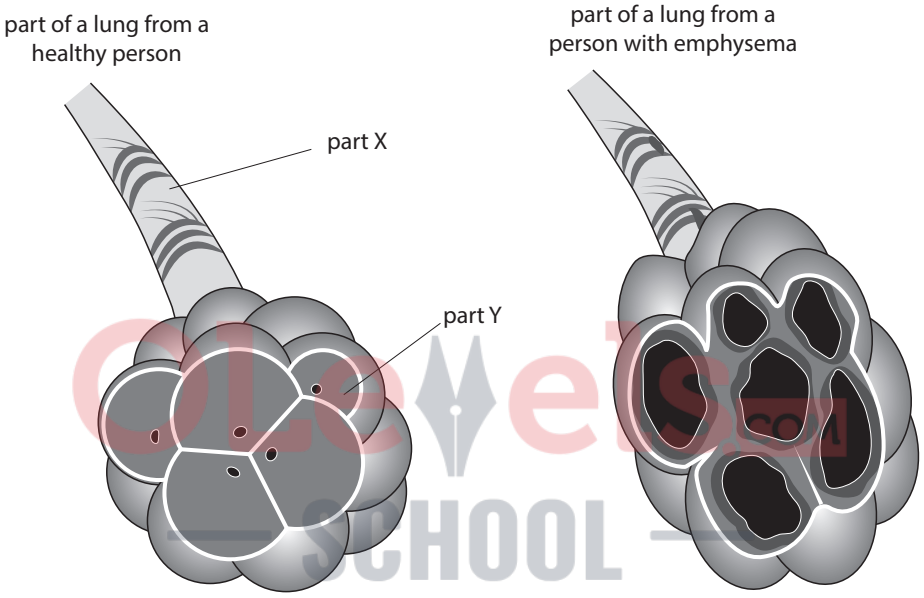
# 7. Question 4 · part b.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.i only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.




(Source: © ttsz / Getty Images)

(i) Name part X.

**(1)**

.....

(ii) Name part Y.

**(1)**

.....

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## 7. Question 4 · part b.i · 1 mark · Working and answer

Working and answer

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# 8. Question 4 · part b.ii · 1 mark

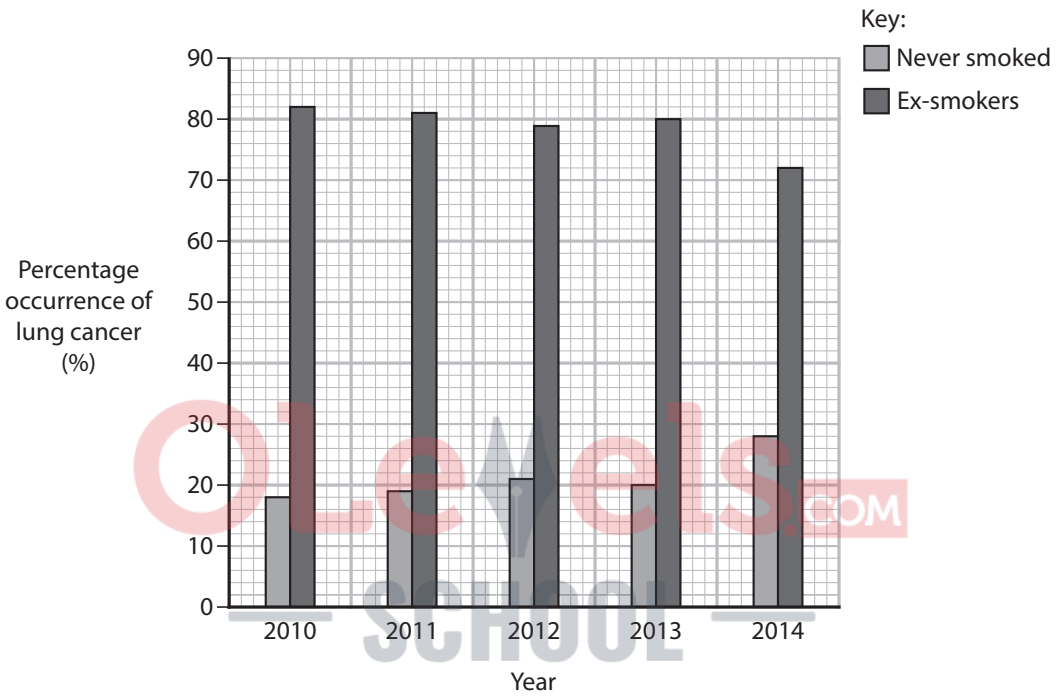
June 2025 Paper 2R

Further practice · Attempt part b.ii only.

**4** (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.




(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

**(2)**

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## 8. Question 4 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

**(2)**

ratio = ..... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

**(2)**

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(iv) Name the substance in cigarette smoke that causes lung cancer.

**(1)**

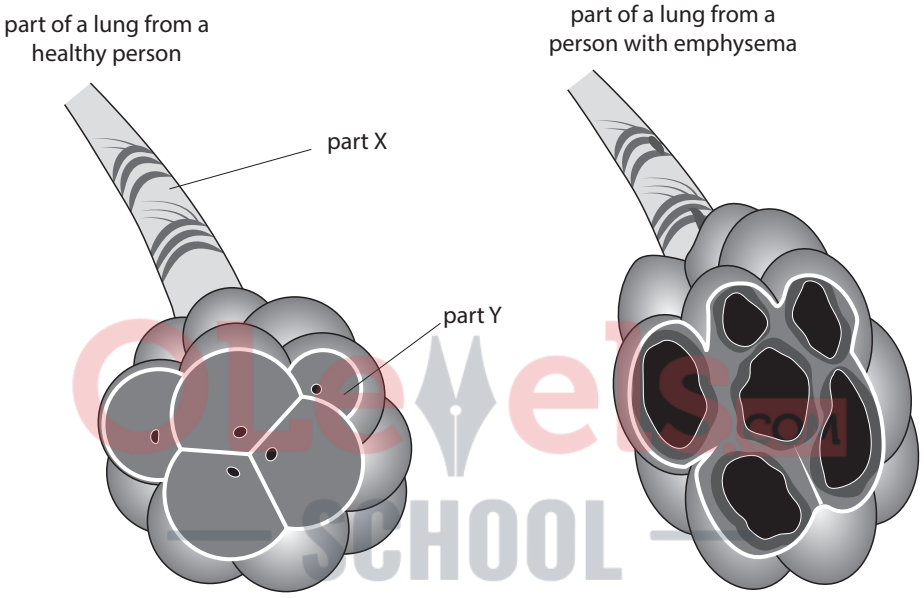
## 8. Question 4 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(b) Emphysema is a lung condition that can be caused by cigarette smoking.

The diagram shows part of a lung from a healthy person and part of a lung from a person with emphysema.




(Source: © ttsz / Getty Images)

(i) Name part X.

**(1)**

(ii) Name part Y.

**(1)**

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## 8. Question 4 · part b.ii · 1 mark

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

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(iii) Explain the effect that emphysema has on the health of a person.

**(4)**

**(Total for Question 4 = 13 marks)**

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Working and answer